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UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Fakulti Komputeran

MID-TERM EXAMINATION SEMESTER 2 2015/2016

COURSE CODE	**:**	**SCSJ 2253**
COURSE NAME	**:**	**REQUIREMENTS ENGINEERING & SOFTWARE MODELING**
YEAR/PROGRAM	**:**	**2 SCSJ**
TIME	**:**	**9 AM (2 HOURS)**
DATE	**:**	**1 APRIL 2016**
VENUE	**:**	**BILIK KULIAH 1-3, FAKULTI KOMPUTERAN**

SECTION A : 5 TRUE/FALSE QUESTIONS (10 MARKS)
The questions map to Course Outcome (CO) 1: *Understand the context of requirements engineering in the overall software development process.*

SECTION B : 3 STRUCTURED QUESTIONS (45 MARKS)
The questions map to Course Outcome (CO) 1: *Understand the context of requirements engineering in the overall software development process.*
and Course Outcome (CO) 2: *Demonstrate the ability to write functional and non-functional requirements for different types of systems using formal and informal techniques.*

SECTION C : 1 PROBLEM SOLVING (35 MARKS)
The questions map to Course Outcome (CO) 2: *Demonstrate the ability to write functional and non-functional requirements for different types of systems using formal and informal techniques.*

TOTAL (90 MARKS)

INSTRUCTION: ANSWER ALL QUESTIONS IN THIS BOOKLET

Student Name	
Matric No.	
Section	
Lecturer Name	

THE BOOKLET CONSISTS OF FOURTEEN (14) PAGES INCLUDING THIS COVER PAGE.

SECTION A: TRUE/FALSE **[10 MARKS]**

Instruction: Determine whether the following statements are TRUE [T] or FALSE [F]. Give reason to justify the FALSE answer.

Question	T/F	Reason
1. The following statement is related to the quality requirement: *"An organizational or technological requirement which restricts the way the system shall be developed"*		
2. The irrelevant environment needs to be considered when defining the system boundary and the context boundary to understand the requirements for the system to be developed.		
3. In certain period of system development life cycle based on Kano Model, the delighters factor of stakeholders requirements may not turn into satisfiers factor and finally into dissatisfiers factor.		
4. The following statements are appropriate for sources of requirements: (a) Standards, norms or laws documents (b) Exclusive stakeholders (c) Other IT systems		
5. In the functional perspective, requirements for a system are documented in terms of: (a) Structure (input and output data) (b) Static aspects of usage and dependency relations (c) Context and its boundaries		

SECTION B: STRUCTURED QUESTION [45 MARKS]

Instruction: Answer **ALL** questions in this section in the provided spaces.

Question 1: [15 marks]

(a) Define the following terms: (6 marks)

(i) Stakeholders

(ii) Requirements Elicitation

(iii) Functional Requirements

(b) Based on the following statement:

> "Organizations are different, projects are different, and cultures are different, thus; Requirements Engineering practices that work in one situation might not be just right for another. Apart from that, knowledge, skills, and experience are the key factors that contribute into creating a well-rounded RA. Additionally, a good RA plays as a mediator between the client and his development team members."

(i) How can Requirements Engineering help in the success of a software project? (2 marks)

(ii) Describe ONE Requirements Analyst (RA) role in defining and understanding the requirements for a new or enhanced software system. (2 marks)

Answer:

(c) Analyze the following set of initial requirements for the proposed University Teaching Support System in Table 1. Classify the requirements based on the 3 types: **Functional [F], Quality [Q] or constraints [C**]. (5 marks)

Table 1: Requirements for University Teaching Support System

University Teaching Support System	
Requirements	**Types [F | Q | C]**
(i) The system shall be delivered and ready for operation at the beginning of the next Semester 1, 2017/2018 sessions.	
(ii) Students shall be able to perform the exam registration process in less than 5 minutes.	
(iii) The system shall be available between 5 am and 3 am of the following day on weekdays, and from 6 am until 11 pm during the weekend.	
(iv) The system shall generate an examination certificate automatically once a lecturer has entered the exam grades.	
(v) The format for storing examination results shall be compatible to the one employed in TeachSupportV2.5 policies document.	

Question 2: [10 marks]

Based on the following descriptions:

Flight Booking Web Portal

First of all, the customers should be able to search for flights (with return or one-way) from and to a desired airport, as well as the depart date and the return date. It should be possible for them to select the desired airline they want to travel with, the option of selecting a non-stop flight, the fare types economy, business or first class, and the number of passengers that would come along. The flight booking web portal will then list the search matching flights, which can be sorted by price, departure time, and airline.

After selecting the desired flights, it shall be bookable. By booking the flight, the customer has to enter his personal data, like his name, gender date of birth, email address, and phone number. Thereafter, it shall be possible for the customer to pay the flight tickets with his credit card, PayPal account or per bank transfer.

After the booking is completed, the confirmation, travelling plan, flight tickets and the site plan of both airports would be electronically accessible on the website and sent to the customer per snail mail or email, as might be desired. If the flight has one or more stopovers, information about entering the country and transfer regulations should be accessible. Additionally, if the layover is longer than one day, a hotel should be bookable by the use of an external hotel booking web portal, which should be recommended by the flight-booking portal. The passenger's luggage should be insurable by an external insurance portal that is recommended by the flight portal. After the flight, the passengers should be able to rate the airline.

(a) Identify FIVE context objects. (5 marks)

(b) Identify a grey zone, or, a grey zone object, and provide a reason to your answer. (5 marks)

Question 3: [20 marks]

1. Based on the following statement, answer the following questions (a) and (b).

Your company is targeting to produce an Internet TV that shall be introduced in the year 2020.

(a) Your team would like to elicit requirements from the mass market to help in the production of the Internet TV software. Suggest TWO most suitable techniques to perform the elicitation and justify why. (4 marks)

(b) Your company then would like to classify system features using the Kano model. Categorise all the following features in Table 2 according to the Kano classification schema: Delighter, Performance, Basic. (10 marks)

Table 2: Features for Internet TV 2020

Features	**Kano classification**
F1: The TV shall be able to show all mainstream channels	
F2: The TV shall enable the user to record a program to watch later	
F3: The TV shall be able to enable the user to search what they want to see	
F4: The TV shall be able to use for video conferencing	
F5: The TV shall be able to allow user to type in information	
F6: The TV shall be able to recommend channels or programs based on the user interests	
F7: The TV shall be able to show information on what is currently showing	
F8: The TV shall enable the user to choose subtitle from multiple languages	
F9: The TV shall be able to turn to standby mode when no one is watching	
F10: The TV shall be able to work with other devices, such as USB, smartphones, computers	

2. Describe the following Goal Model (Figure 1). (6 marks)

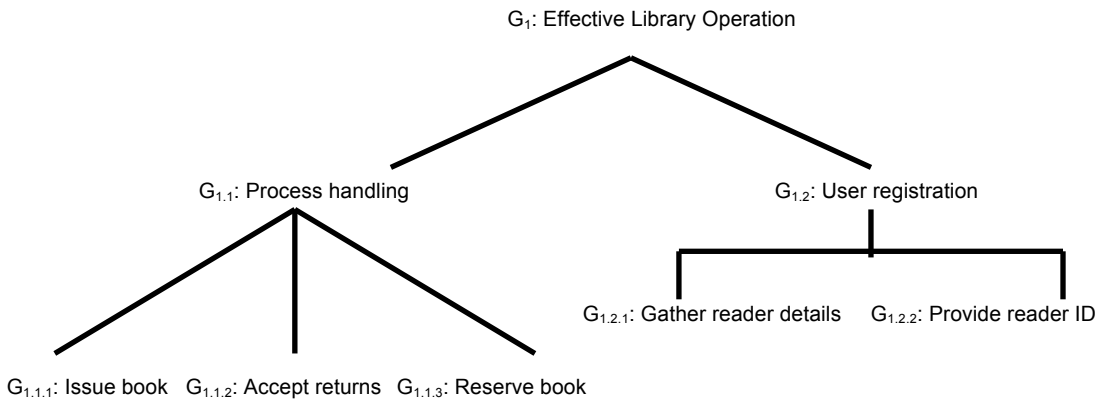


Figure 1. Library Operation Goal Model

Answer:

SECTION C: PROBLEM SOLVING [35 MARKS]

Instruction: Answer **ALL FOUR** questions in the provided spaces.

Below is a case study scenario of Q-Taxi system.

Q-Taxi

Q-Taxi is a small independent taxi company operating in a major city. Q-Taxi owns 25 taxis, which it rents out to drivers on an annual basis. Each vehicle is effectively rented out to three drivers to cover three 8-hour shifts in a day: therefore there are 75 taxi drivers contracted to Q-Taxi at any given time.

Q-Taxi is a profitable company because it has built up a good reputation locally, and there is always a waiting list of drivers wanting to apply to rent a vehicle. Each driver pays an annual rental fee in advance to Q-Taxi, giving him or her use of a vehicle for 8 hours a day every day of the year. In addition to the annual rental, Q-Taxi takes 5% of the money a driver earns every week.

Q-Taxi is responsible for taxing, insuring and maintaining the vehicles. If a vehicle is due for a service or needs to be repaired Q-Taxi contacts a garage and arranges it. Q-Taxi keeps an account of the repair and service costs for each vehicle.

At the end of each shift drivers give the money they have earned to Q-Taxi. If they needed to refuel the vehicle they also submit an expense claim at the end of the shift. At the end of every week Q-Taxi calculates the amount owing to each driver based on the money earned from fares, the expense claims and the deduction of 5%. The drivers are then paid.

Q-Taxi intends to develop a computerized system to handle its business operation. You are required to (see the following pages):

Question 1: [10 Marks]

Produce a **Swimlane diagram** to represent the ***as-is*** manual business activities and processes of Q-Taxi.

Answer:

Question 2: [10 Marks]

One of the possible Q-Taxi business process that could be automated is the "Arrange Taxi Services and Repair". The automated process must cover from the process of the manager of Q-Taxi enters the details of vehicle needs to be repaired or serviced to the process of the garage company claim the cost for the repair or service.

Produce a **Swimlane diagram** to represent the ***to-be*** automated business process for the "**Arrange Taxi Services and Repair**".

Answer:

Question 3: [5 marks]

Produce a **Use Case diagram** for the computerized Q-Taxi system.

Answer:

Question 4: [10 marks]

Produce a complete use case specification for the "Arrange Taxi Services and Repair" use case.

Answer:

Use Case Name	
Use Case ID	
Description	
Actors	
Pre-conditions	
Normal Flow	
Alternative Flow	
Exception	
Post-conditions	

[[Ends]]

<<Extra Paper>>

<<Extra Paper>>